Jerry Zhou

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EDUCATION

University of California, Berkeley | GPA: 4.0/4.0 **May 2029**

- BS in Electrical Engineering and Computer Science, College of Engineering
- Relevant Coursework: Data Structures and Algorithms, Signals and Information Processing, Physics: Mechanics, Physics: Electricity and Magnetism, Linear Algebra, Multivariable Calculus

TECHNICAL SKILLS

Software: KiCAD, Altium, Fusion 360, Solidworks, Java, Scheme, SQL, Python, C/C++, HTML, CSS, MATLAB
Tools & Frameworks: Bullet, MuJoCo, CoppeliaSim, MoveIt, ROS2, Git, Pandas, PyTorch, Tensorflow, JAX

EXPERIENCE

Biomimetic Millisystems Lab **Berkeley, CA**
Undergraduate Researcher supervised by Dr. Ronald Fearing & Dr. John Atkins *October 2025-Present*

- Developed **physics-based system identification models** for ultrasonic actuators by validating dynamic models from experimental data and integrating them into **CoppeliaSim** for a **MRI-compatible 7-DOF robotic system**
- Implemented **backlash and hysteresis models** in Python using **quasi-linear and logarithmic fits** in Python to enable reliable **parameter extraction** from experimental measurements and improve model fidelity
- Improved closed-loop tracking robustness by designing and tuning **PD and impedance controllers with disturbance observers** to maintain performance under varying load conditions

Formula Electric at Berkeley **Berkeley, CA**
Accumulator Electrical Engineer *September 2025-Present*

- Designed and architected a **four-layer high-voltage BMS PCB** using **Altium Designer** to support a 1403p accumulator with fault-tolerant voltage/temperature sensing, active balancing, and galvanic isolation
- Integrated and debugged BMS hardware by validating **isoSPI communication**, ADBMS voltage sensing, and **STM32 balancing firmware** to ensure reliable operation under high-voltage conditions
- Verified **system-level functionality** through bench testing, resistor-divider emulation, and iterative bring up to identify and resolve **signal integrity** and sensing issues prior to deployment

Innovative Building Energy Control **Lake Forest, CA**
Test and Validation Engineer: *December 2025-January 2026*

- Commissioned and validated **50+ isolated AC-DC PCBs** by performing power-up, regulation stability analysis, and **ATmega328p** bring-up tests to verify production-ready electrical and functional performance
- Diagnosed and resolved power integrity and control issues using **oscilloscopes and multimeters** to analyze voltage regulation, startup behavior, and Atmega328P **I/O and clock signals** to ensure stable power delivery
- Implemented **solar panel telemetry acquisition**, displaying real-time data on LCDs and transmitting via **Wi-Fi**

Magnitude.io **Berkeley, CA**
Electrical Engineering Consultant: *September 2025-December 2025*

- Designed and built a **random positioning machine** to simulate microgravity conditions under a **$400 BOM**, generating randomized motion profiles using **Fourier-series-based and PRBS trajectories**
- Schematized and brought up **a custom two-layer control PCB** integrating an ESP32, motor drivers, sensors, and display interfaces (I2C, SPI, Bluetooth), with **24V-to-12/5/3.3V power distribution** for operation under load

PROJECTS

Formula One Prediction Model | *Pandas, LightGBM, Python, Weather/Kaggle API* ***January 2026***

- Constructed a **learning-to-rank model** for F1 race prediction with **gradient-boosted decision trees** achieving **NDCG@1/3/10** of **0.78/0.77/0.83**, processing **25000+ driver-race combinations** (1950-2020)

Scheme Interpreter | *Python* ***November 2025***

- Implemented a full Scheme interpreter in Python, featuring expression parsing, lexical scoping via environment chaining, user-defined procedures, and an extensible primitive system